An-Najah Nation University

Faculty of Engineering

جامعة النجاح الوطنية كلية الهندسة

Computer Engineering Department

Networks1 (10636454)

Programming HW project

ChatLite — Minimal Real-Time Chat System using Java Sockets

Instructor name:
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Student#1 Name:……

Student#2 Name:……

Academic Year: **2025-2026**

Registration Number#1: ……..

Semester: **Spring**

Registration Number#2: ……..

Credit Hours: **3**

Date: **01/April/2026**

Total Exam Mark: **15**

Exam Weight: **15%**

Deadline: 31/March/2026 midnight

Question	Points	ILO's	ILO's%	Question Grade	Required Time
Q1	15				2 Weeks
Student Grade					

Exam Notes:

1- Teamwork, groups of two.
2- marks may vary between group members based on the discussion session.

1. Introduction

This document describes the ChatLite programming project. The goal of this project is to implement a minimal real-time chat system using Java sockets and a graphical user interface. Students will implement a multi-client server and a GUI client capable of real-time messaging, chat rooms, and private communication.

2. Learning Goals

- Understand TCP client/server programming using Java sockets.
- Design and parse a simple text-based communication protocol.
- Implement concurrent servers using Threads or ExecutorService.
- Develop real-time communication applications.
- Design simple Java GUI interfaces and handle user events.

3. System Overview

The ChatLite system consists of two main components:

- ChatLite Server:

  Responsible for managing connected users, chat rooms, message broadcasting and private messaging. Each client connects using a persistent TCP connection.

- ChatLite Client:

  A Java GUI application that allows users to login, join chat rooms, send messages, view online users and send private messages.

4. Communication Protocol

The system uses a simple text-based protocol over TCP connections. Each command is transmitted as a line of ASCII text terminated by a newline character.

Example session:

```
C: HELLO <username>
S: 200 WELCOME

C: JOIN <room>
S: 210 JOINED <room>

C: MSG <room> <message>
S: 211 SENT
```

C: PM <username> <message>
S: 212 PRIVATE SENT
C: USERS
S: 213 <count>
S: 213U <username>
... repeated ...
S: 213 END
C: ROOMS
S: 214 <room1>
S: 214 <room2>
C: LEAVE <room>
S: 215 LEFT
C: QUIT
S: 221 BYE

5. Client Graphical User Interface

The client interface should include the following components:

- Chat Rooms list displaying available rooms.
- Online Users panel showing connected users and their status.
- Chat Messages display area with username and timestamp.
- Message input field and Send button.
- Private messaging panel for direct communication.
- Search box for filtering messages.
- Connection status bar displaying system logs.

Figure 1 shows the suggested layout for the ChatLite Client GUI.

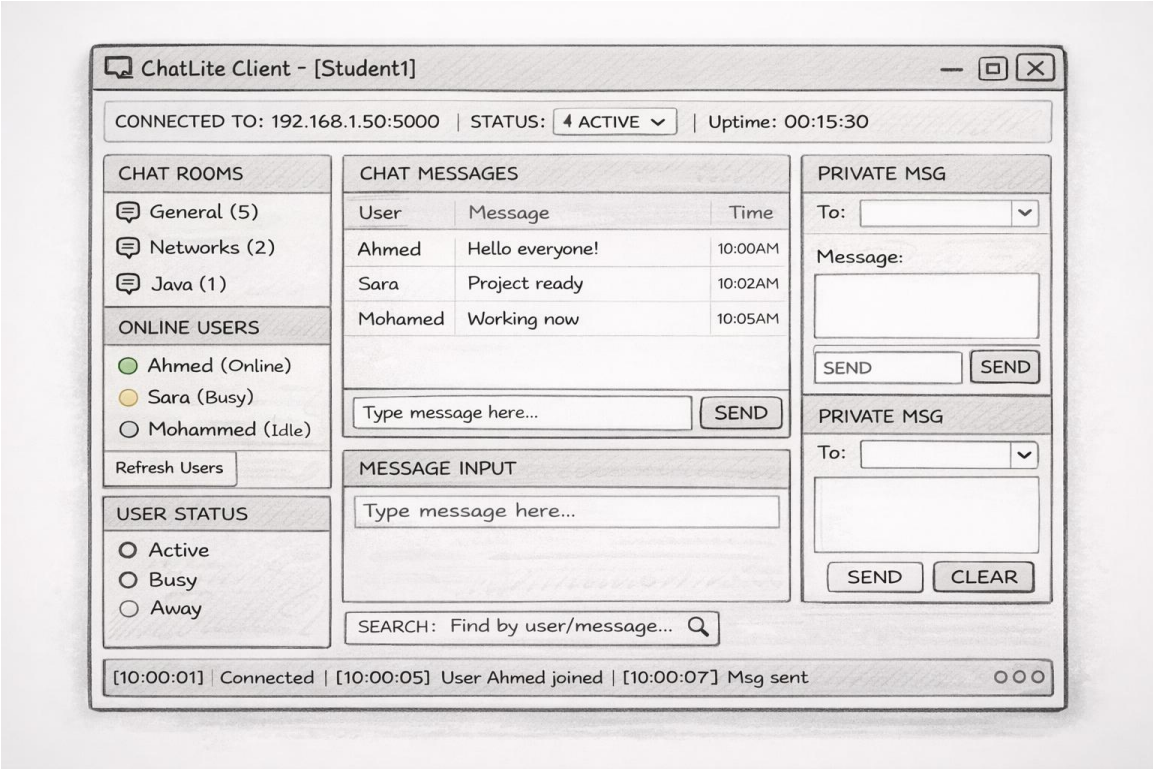

Figure 1: ChatLite Client GUI Layout

6. Server Console Interface

The server console provides monitoring and administrative controls for the system.

- Server status indicator showing ONLINE state and active ports.
- User management panel for creating and managing users.
- Active sessions list showing usernames and IP addresses.
- Mailbox statistics displaying sent and received messages.
- Broadcast message capability.
- Real-time system logs showing events and errors.
- Configuration options such as maximum message size.

Figure 2 shows the suggested layout for the ChatLite Server Console.

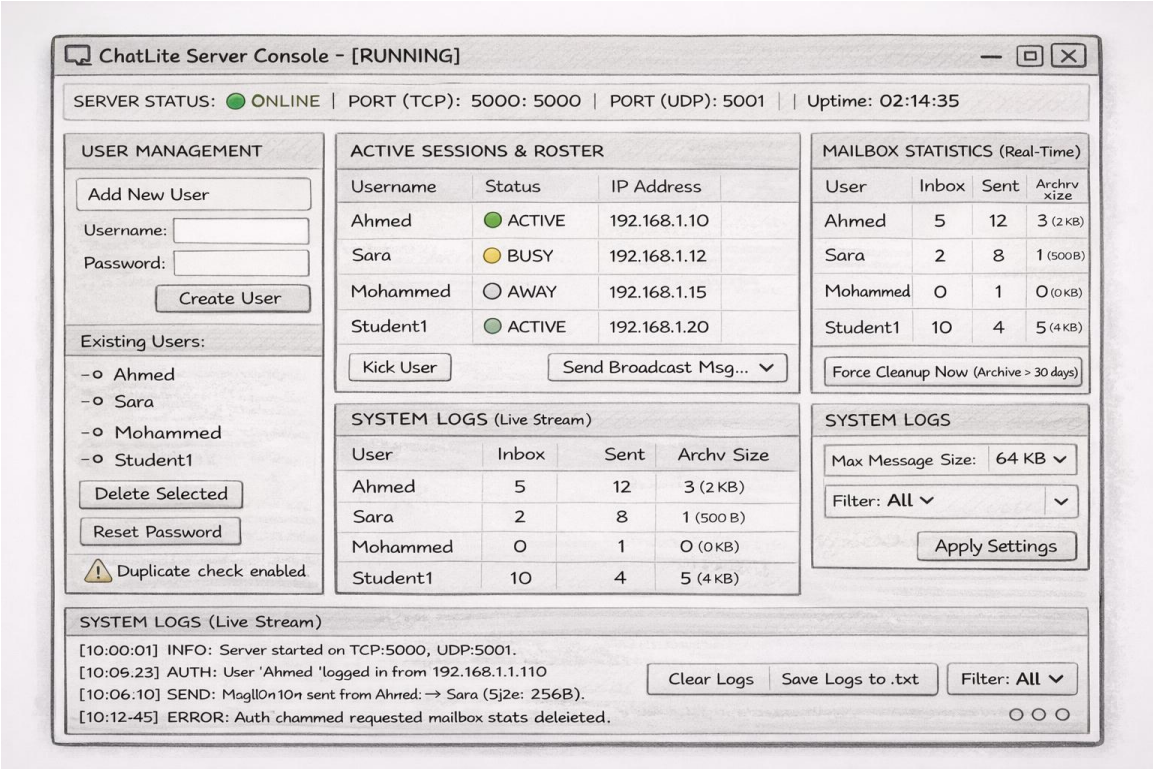

Figure 2: ChatLite Server Console GUI

7. Project Deliverables

- Source code for both Client and Server.
- README file explaining how to compile and run the system.
- A short report (3–5 pages) describing architecture, protocol, GUI design and testing.
- A short demo video showing server start, two clients connecting, joining rooms and exchanging messages.
- Sample users and rooms used for testing.